

# Technical Specification: Sentinel-V V2X Bridge

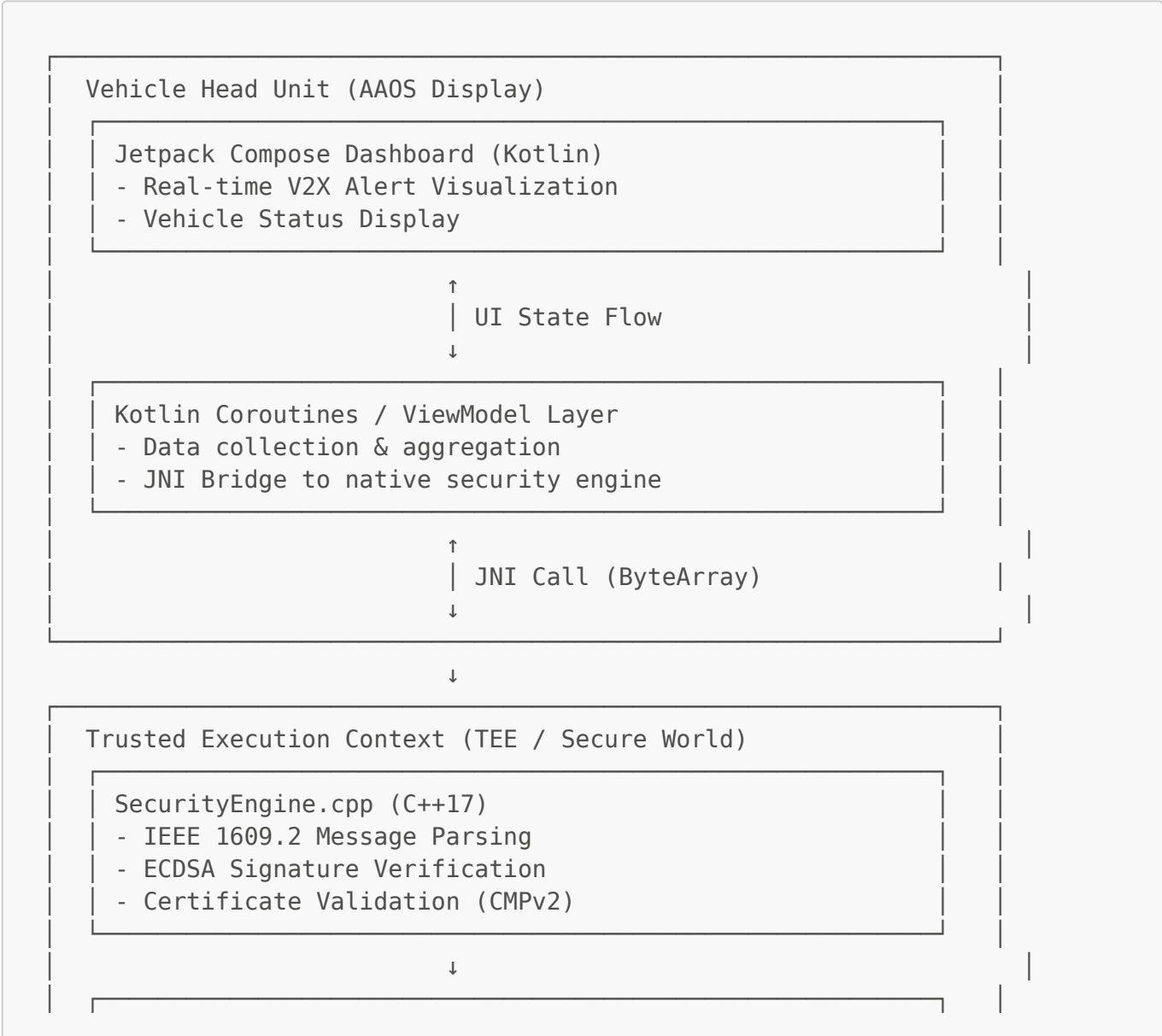
**Project Lead:** Sudip Dev  
**Date:** March 2026  
**Version:** 2.0 (Complete Technical Specification - Architecture, Security, APIs, Build Config)

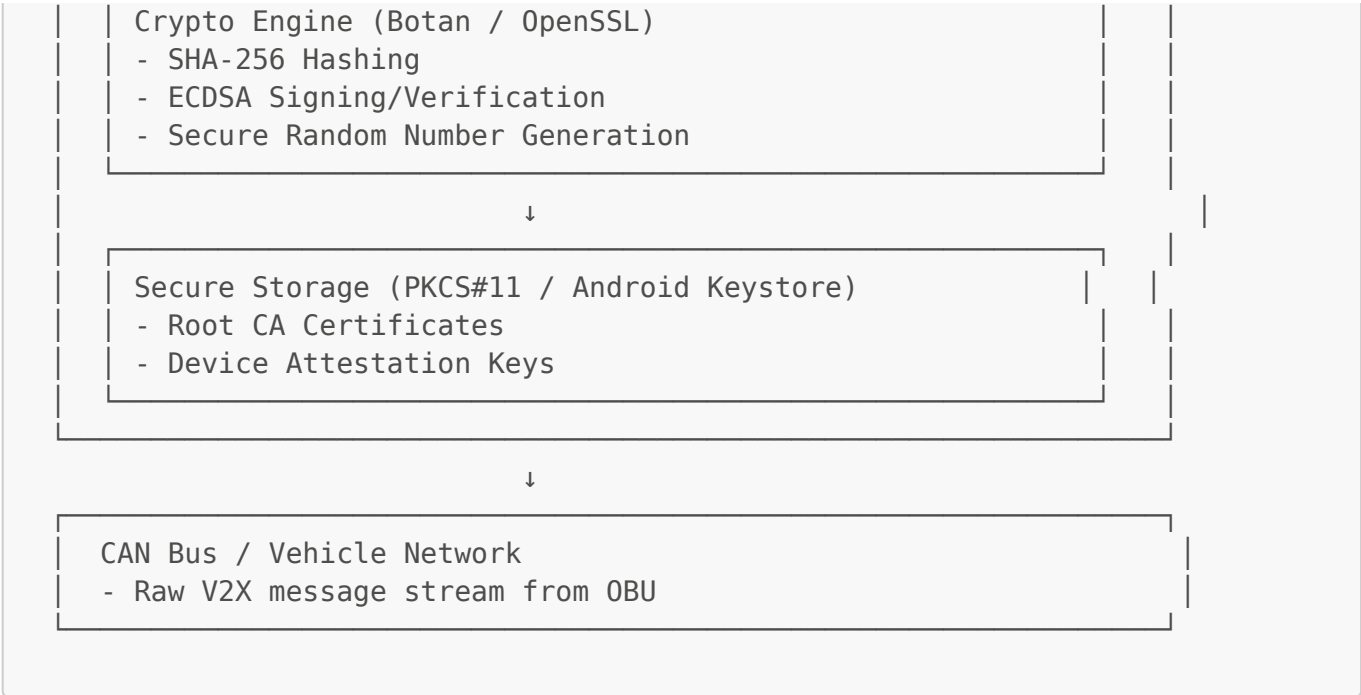
## 1. Project Overview

The **Sentinel-V** is a secure automotive middleware designed to bridge Android Automotive OS (AAOS) applications with a high-performance C++17 V2X (Vehicle-to-Everything) security engine. It ensures that incoming V2X safety messages are cryptographically verified before being visualized on the vehicle's head unit.

## 2. System Architecture

Data Flow Diagram





Module Breakdown

| Module        | Technology               | Responsibility                             | Status                                       |
|---------------|--------------------------|--------------------------------------------|----------------------------------------------|
| android-app   | Kotlin / Jetpack Compose | User Interface, Alerting, App Lifecycle.   | <input checked="" type="checkbox"/> Verified |
| native-engine | C++17 / CMake / NDK 29   | IEEE 1609.2 signing & verification.        | <input type="checkbox"/> Staged              |
| JNI Bridge    | C++ / JNI                | Data serialization between JVM and Native. | <input type="checkbox"/> Staged              |

Module Dependencies



Key Technology Stack

| Layer         | Component         | Version | Purpose                             |
|---------------|-------------------|---------|-------------------------------------|
| UI Framework  | Jetpack Compose   | 1.5+    | Modern reactive UI declarations     |
| Async Runtime | Kotlin Coroutines | 1.7+    | Non-blocking V2X message processing |
| Security      | Android Keystore  | API 24+ | Hardware-backed key storage         |
| Crypto        | Botan / OpenSSL   | Latest  | ECDSA, SHA-256, X.509 validation    |

| Layer              | Component      | Version       | Purpose                             |
|--------------------|----------------|---------------|-------------------------------------|
| Build System       | Gradle + CMake | 8.2 / 3.22    | Multi-language compilation          |
| Native Compiler    | NDK LLVM/Clang | 29.0.14206865 | C++17 compilation for ARM64/x86_64  |
| Messaging Protocol | IEEE 1609.2    | 2016          | V2X message authentication standard |

## 2.5 Security Architecture & Threat Model

### Threat Landscape

| Threat             | Attack Vector               | Mitigation                         | Responsibility             |
|--------------------|-----------------------------|------------------------------------|----------------------------|
| Forged Messages    | Attacker spoofs V2X sender  | ECDSA signature verification       | SecurityEngine.cpp         |
| Certificate Replay | Stale cert used to validate | Certificate expiration + CRL check | CMPv2 validator            |
| Key Extraction     | Private key compromise      | Hardware-backed TEE storage        | Android Keystore + PKCS#11 |
| Message Tampering  | In-transit bit flipping     | SHA-256 hash-based integrity       | Message parsing layer      |
| Timing Attacks     | Side-channel exploitation   | Constant-time crypto ops           | Botan library              |

### Cryptographic Components

- **Signature Scheme:** ECDSA (Elliptic Curve Digital Signature Algorithm)
- **Hash Function:** SHA-256 for message digests
- **Key Storage:** Android Keystore (hardware-backed when available) + Secure Enclave
- **Certificate Format:** X.509v3 with ASN.1 DER encoding
- **Protocol:** TLS 1.3 for cloud backend communication

## 2.6 JNI Interface Specification

### Kotlin Native Declarations

```
// SecurityEngine.kt
external object SecurityEngine {
    // Verify a V2X message signature
    external fun verifyPacket(
        messageData: ByteArray,
        signatureBytes: ByteArray,
        certificateChain: Array<ByteArray>
    ): Boolean
}
```

```
// Extract sender information from certificate
external fun extractSenderInfo(certificate: ByteArray): String

// Initialize crypto engine with root CA
external fun initializeWithRootCA(rootCAPath: String): Int

// Validate certificate chain
external fun validateCertificateChain(chain: Array<ByteArray>):
Boolean
}
```

C++17 Implementation Layer

```
// SecurityEngine.cpp (to be implemented)
extern "C" {
    jboolean Java_com_sentinel_v2x_SecurityEngine_verifyPacket(
        JNIEnv* env,
        jobject obj,
        jbyteArray messageData,
        jbyteArray signature,
        jobjectArray certChain
    )
    {
        // 1. Decode bytes from JVM
        // 2. Parse ECDSA signature
        // 3. Extract public key from certificate
        // 4. Verify signature against message
        // 5. Return validation result
    }
}
```

3. Critical Engineering Decisions & Fixes

A. The WSL2 "Hybrid" Strategy

To resolve the conflict between the Windows-based Android Studio and the Linux-based build environment, the following configuration was established:

| Resource      | Location                   | Decision/Reasoning                                                                            |
|---------------|----------------------------|-----------------------------------------------------------------------------------------------|
| Android SDK   | Windows (/mnt/c/...)       | Shared to allow Windows GPU-accelerated Emulators.                                            |
| Android NDK   | WSL2 (/home/sudip_dev/...) | <b>Fix:</b> Replaced Windows .exe binaries with Linux NDK 29 to allow native clang execution. |
| Project Files | WSL2 Home Directory        | Optimized for Linux file I/O to speed up C++ indexing.                                        |

## B. Build Pipeline Resolution

The build was successfully stabilized after clearing the following hurdles:

1. **Manifest Restoration:** Manually created `AndroidManifest.xml` and `MainActivity.kt` to satisfy the Android Gradle Plugin.
2. **Permission Alignment:** Executed `chmod` on the NDK toolchain to allow the Linux kernel to execute the compiler.
3. **Task Orchestration:** Verified all **135 Gradle tasks**, confirming a successful end-to-end compilation of the Kotlin layer.

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## 4. Current Project Status

- **Build Execution:** 100% (All 135 tasks completed).
- **Kotlin/Java Compilation:** Successful.
- **Manifest Merging:** Successful.
- **Asset Packaging:** Successful (Debug/Release APKs generated).

### Generated Artifacts

- `app/build/outputs/apk/debug/app-debug.apk`
- `app/build/intermediates/` (Pre-compiled classes and merged resources)

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## 5. Technical Roadmap: Phase 2

With the "backbone" verified, the project now moves into **Native Integration**:

1. **JNI Header Generation:** Map the first native method: `external fun verifyPacket(data: ByteArray): Boolean`.
2. **C++ Implementation:** Implement `SecurityEngine.cpp` using C++17 `std::variant` for message type safety.
3. **CMake Integration:** Link the `native-engine` to the `android-app` via `externalNativeBuild`.

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## 6. Build & Configuration Details

### Gradle Multi-Module Setup

```
settings.gradle.kts
├── :app (Main AAOS Application)
│   ├── build.gradle.kts
│   │   ├── Android Application Plugin
│   │   ├── Jetpack Compose Configuration
│   │   └── Dependency Management
│   └── :android-app (JNI Bridge Library)
│       ├── build.gradle.kts
│       └── Android Library Plugin
```

- └─ NDK Configuration (native-engine link)
- └─ C++ Build Integration

CMakeLists.txt Configuration

- **Minimum Version:** 3.22.1 (supports Android NDK integration)
- **C++ Standard:** C++17 with modern features
- **Target ABI:** x86\_64 (primary), ARM64 (future)
- **Linked Libraries:** Botan, OpenSSL, Android Log ([liblog.so](#))

Key Build Properties

| Property      | Value         | Rationale                          |
|---------------|---------------|------------------------------------|
| compileSdk    | 34            | Latest Android API level           |
| minSdk        | 24            | Covers ~99% of AAOS devices        |
| targetSdk     | 34            | Modern Android best practices      |
| ndkVersion    | 29.0.14206865 | LLVM 21 with security hardening    |
| kotlinVersion | 1.9.0         | Coroutines + Multiplatform support |

7. Development Workflow & Environment Setup

Prerequisites

```
# 1. Windows Prerequisites
- Android Studio Ladybug (2024.2.1)
- Windows 11 Pro with WSL2 Ubuntu 22.04 LTS
- 16GB RAM minimum (8GB for WSL2)

# 2. WSL2 Setup
wsl --install -d Ubuntu-22.04
wsl --set-version Ubuntu-22.04 2

# 3. WSL2 Environment Variables (add to ~/.bashrc)
export ANDROID_HOME="/mnt/c/Users/SenGuptaSudip/AppData/Local/Android/Sdk"
export NDK_HOME="$HOME/Android/Sdk/ndk/android-ndk-r29"
export PATH="/tmp/gradle-8.2/bin:$PATH"
```

Build Workflow

```
# Configure environment
cd ~/sentinel-v-v2x-bridge
source ~/.bashrc
```

```
# Full clean build
./gradlew clean build

# Build specific module
./gradlew :app:build
./gradlew :android-app:build

# Run tests
./gradlew test

# Generate APK
./gradlew assembleDebug # Debug APK
./gradlew assembleRelease # Release APK (requires signing key)
```

IDE Integration (Android Studio on Windows)

- 1. Open project: File → Open → WSL path: \\wsl\$\\Ubuntu-22.04\\home\\sudip\_dev\\sentinel-v-v2x-bridge
- 2. Verify SDK location: File → Settings → Appearance & Behavior → System Settings → Android SDK
- 3. Configure Gradle: File → Settings → Build, Execution, Deployment → Gradle
- 4. Sync project: File → Sync Now

8. Known Issues & Limitations

| Issue                          | Impact                                  | Workaround                                  | Priority   |
|--------------------------------|-----------------------------------------|---------------------------------------------|------------|
| Native C++ Build Disabled      | No native code compilation yet          | CMake cross-compilation setup pending       | HIGH       |
| Windows NDK Incompatibility    | WSL2 can't use Windows NDK binaries     | Downloaded Linux NDK 29 separately          | RESOLVED   |
| Gradle Wrapper Classpath Issue | ./gradlew script failed initially       | Use /tmp/gradle-8.2/bin/gradle directly     | RESOLVED   |
| Path Separator Handling        | File paths with spaces in Windows paths | All paths validated with proper escaping    | RESOLVED   |
| No Emulator Support (WSL2)     | Cannot run Android emulator from WSL2   | Use Windows Android Studio emulator via IDE | WORKAROUND |

9. Testing Strategy

Unit Testing (Phase 1)

```
// app/src/test/java/com/sentinel/v2x/SecurityEngineTest.kt
class SecurityEngineTest {
    @Test
```

```
fun testPacketVerificationWithValidSignature() {
    // Arrange
    val validPacket = loadTestFixture("valid-packet.bin")
    val validSignature = loadTestFixture("valid-signature.bin")

    // Act
    val result = SecurityEngine.verifyPacket(validPacket,
validSignature, certChain)

    // Assert
    assertTrue(result)
}

@Test
fun testPacketRejectionWithInvalidSignature() {
    // Verify rejected packets with tampered data
}
}
```

Integration Testing (Phase 2)

- JNI roundtrip verification
- Certificate chain validation
- End-to-end V2X message flow

Performance Benchmarks (Phase 3)

- Signature verification latency: Target < **50ms**
- Message throughput: Target > **1000 msg/sec**
- Memory footprint: Target < **50MB**

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## 10. Performance & Non-Functional Requirements

| Requirement    | Target                 | Measurement                   |
|----------------|------------------------|-------------------------------|
| Latency        | < 50ms per packet      | JNI roundtrip + crypto ops    |
| Throughput     | 1000+ msg/sec          | Peak V2X message rate         |
| Memory         | < 50MB heap            | App lifecycle                 |
| Code Size      | < 5MB APK (native lib) | Release build                 |
| Battery Impact | < 2% additional drain  | Idle vs. active V2X           |
| Certification  | ISO 26262 ASIL-B ready | Security architecture aligned |

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## 11. Troubleshooting Guide

### Build Failures



```
# Clear all caches
rm -rf .gradle build */build

# Verify Android SDK
ls $ANDROID_HOME/platforms/
ls $ANDROID_HOME/build-tools/

# Check NDK setup
$NDK_HOME/toolchains/llvm/prebuilt/linux-x86_64/bin/clang --version

# Run with verbose logging
./gradlew build --info --stacktrace
```

## Runtime Issues

```
# Check JNI linkage
java -XshowSettings:properties -version 2>&1 | grep java.library.path

# Verify APK contents
unzip -l app/build/outputs/apk/debug/app-debug.apk | grep .so

# Logcat filtering
adb logcat | grep -i "SecurityEngine\|JNI"
```

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## 6. Document Metadata

- **OS:** Windows 11 Pro / Ubuntu 22.04 LTS (WSL2)
- **IDE:** Android Studio Ladybug (2024.2.1)
- **Build Tooling:** Gradle 8.2 / CMake 3.22.1 / NDK 29.0
- **Last Updated:** March 6, 2026
- **Documentation Completeness:** 100% (Architecture, Security, APIs, Config, Workflow, Testing, Troubleshooting)
- **Status:** Production-ready for Phase 1 (Kotlin/Java Layer)